



Postfertilization Autophagy of Sperm Organelles Prevents Paternal Mitochondrial DNA Transmission

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mutant hermaphrodites were mated with wild-type males (Fig. 4B and fig. S7A). Conversely, *lgg-1(tm3489)* mutation in males did not affect clearance of paternal mitochondria in embryos, suggesting that maternally provided autophagic activity is required for effective degradation of sperm-derived mitochondria in the early embryo (fig. S7B). On the other hand, even in embryos from the *lgg-1(tm3489)* hermaphrodites, paternal mitochondria were observed as late as at the lima-bean stage but eventually disappeared at later stages. We assumed that the wild-type *lgg-1* gene provided by the sperm was expressed and may have restored the autophagic activity in late embryos. Supporting this possibility, crossing of *lgg-1(tm3489)* hermaphrodites and wild-type males resulted in viable F1 embryos (fig. S6). To generate *lgg-1(tm3489)* homozygous embryos, MT-labeled *lgg-1(tm3489)* males were crossed with *lgg-1(tm3489)* hermaphrodites. In *lgg-1(tm3489)* homozygous F1 embryos, MT-labeled paternal mitochondria were detected in late embryos and even in L1 larvae, showing that autophagy is required for elimination of paternal mitochondria (Fig. 4C and fig. S8, A to C). The same results were obtained by using sperm-derived mtGFP (fig. S7, C to F'). In mutant larvae, MT-labeled mitochondria were distributed throughout various tissues. To further detect traces of paternal mtDNA in F1 offspring, we used a large deletion allele of mtDNA (*uaDf5*) (Fig. 4D). In hermaphrodites, *uaDf5* is stably inherited in a heretoplasmic state with the wild-type mtDNA (24). When paternal mtDNA was marked with *uaDf5*, it was not normally transmitted to the next generation (Fig. 4E). In contrast, in the *lgg-1(tm3489)* background, paternally provided *uaDf5* was detected in F1 offspring, demonstrating inheritance of paternal mtDNA (Fig. 4F and fig. S8D).

We also examined whether sperm mitochondria were ubiquitinated as reported in mammals (7). In sperm, no obvious staining was detected by using an antibody against polyubiquitin (fig. S9A). Instead, punctate structures giving a strong polyubiquitination signal appeared around paternal pronuclear DNA in early embryos (fig. S9B). We found that these structures did not directly overlap with paternal mitochondria but colocalized with a marker of membranous organelles (MOs), specialized vesicular structures in sperms (fig. S9D) (25). These ubiquitinated MOs were also engulfed by GFP-LGG-1-positive autophagosomes. MOs and paternal mitochondria seemed to be engulfed together or independently by autophagosomes (fig. S9E).

We have shown that autophagy is essential for the effective elimination of paternal mitochondria. Our results also suggest that autophagic degradation of paternal mitochondria could be the mechanism of maternal inheritance of mtDNA. Autophagosomes induced upon fertilization were localized and appeared to selectively engulf sperm components, which is reminiscent of the selective autophagic degradation of damaged mitochondria

and invading bacteria in mammals (26, 27). Paternal mitochondria remaining in the *lgg-1* mutant retained the characteristic granular morphology and did not increase in number during embryogenesis (figs. S7 and S8), implying that they are proliferation- and fusion-inactive. Such qualitative difference may underlie selective elimination of paternal mitochondria. The paternal mitochondria and/or mtDNA could be heavily damaged by reactive oxygen species before fertilization (28). Selective elimination of paternal mitochondria may prevent the spreading of potentially deleterious mitochondria to the whole population. In mice, autophagy is also up-regulated immediately after fertilization (12), suggesting that fertilization-triggered autophagy is a conserved phenomenon in animals.

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Supporting Online Material

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Materials and Methods

Figs. S1 to S9

Table S1

References (29–39)

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Postfertilization Autophagy of Sperm Organelles Prevents Paternal Mitochondrial DNA Transmission

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In sexual reproduction of most animals, the spermatozoon provides DNA and centrioles, together with some cytoplasm and organelles, to the oocyte that is being fertilized. Paternal mitochondria and their genomes are generally eliminated in the embryo by an unknown degradation mechanism. We show that, upon fertilization, a *Caenorhabditis elegans* spermatozoon triggers the recruitment of autophagosomes within minutes and subsequent paternal mitochondria degradation. Whereas the nematode-specific sperm membranous organelles are ubiquitinated before autophagosome formation, the mitochondria are not. The degradation of both paternal structures and mitochondrial DNA requires an LC3-dependent autophagy. Analysis of fertilized mouse embryos shows the localization of autophagy markers, which suggests that this autophagy event is evolutionarily conserved to prevent both the transmission of paternal mitochondrial DNA to the offspring and the establishment of heteroplasmy.

In most metazoans the mitochondrial DNA (mtDNA) is maternally inherited (1), and typically, individuals harbor a unique mtDNA

genotype defining the homoplasmic state. Heteroplasmy, the occurrence of more than one mtDNA genotype, is rare and results from spontaneous

modifications of maternal mtDNA or paternal contribution upon fertilization (2). In mice and *C. elegans*, heteroplasmy can be inherited through the female germ line but not through spermatozoa (3, 4). Paternal mitochondria and their mtDNA are eliminated, but the mechanism is not yet understood. Studies in mammals have suggested that elimination of paternal mitochondria is dependent on a mechanism involving ubiquitination and the lysosomal pathway (5). Furthermore, medaka sperm mtDNA appears to be progressively degraded 30 to 60 min after fertilization by an unknown mechanism (6).

Autophagy is a major ubiquitous catabolic process in eukaryotes allowing the degradation of cytoplasmic constituents and organelles by selective or nonselective sequestration in double-membrane vesicles, the autophagosomes (7). In the mouse early embryo, autophagy is essential

for preimplantation development (8) and has been postulated to degrade maternal proteins. However, its role in the elimination of paternal inherited components remains elusive. To address this question, we analyze the fate of sperm components in the *C. elegans* embryo (Fig. 1A) (see also supporting online materials and methods). Here, we show that upon fertilization, several sperm components, including mitochondria, enter the oocyte and are rapidly degraded by autophagy.

C. elegans mature spermatozoa contains membranous organelles (MOs) that represent nematode-specific membrane compartments and are present either underneath or fused with the plasma membrane (Fig. 1A) (9). Their function is unclear, but MO fusion is essential for sperm motility (10). *C. elegans* spermatozoa also contain mitochondria and high amounts of the major sperm proteins (MSPs) concentrated in the pseudopod and required for sperm motility (Fig. 1A). To analyze which of these components enter the oocyte upon fertilization, we used antibodies specific for the MOs or the MSPs and the inner mitochondrial membrane adenine nucleotide transporter ANT-1.1 fused to green fluorescent protein (GFP) (11) or CMXRos mitotracker to label mitochondria. After crossing labeled males with wild-type (WT) hermaphrodites, we analyzed the presence of MOs, MSPs, mitochondria, and sperm nuclear DNA in the progeny (Fig. 1 and fig. S2, A and B). After fertilization, both MOs and sperm mitochondria enter and remain close to the sperm

DNA at the posterior pole of the embryo (Fig. 1, C and D). MSPs are uniformly distributed in newly fertilized embryos up to the two-cell stage (fig. S2A) with no enrichment around sperm DNA (fig. S2B).

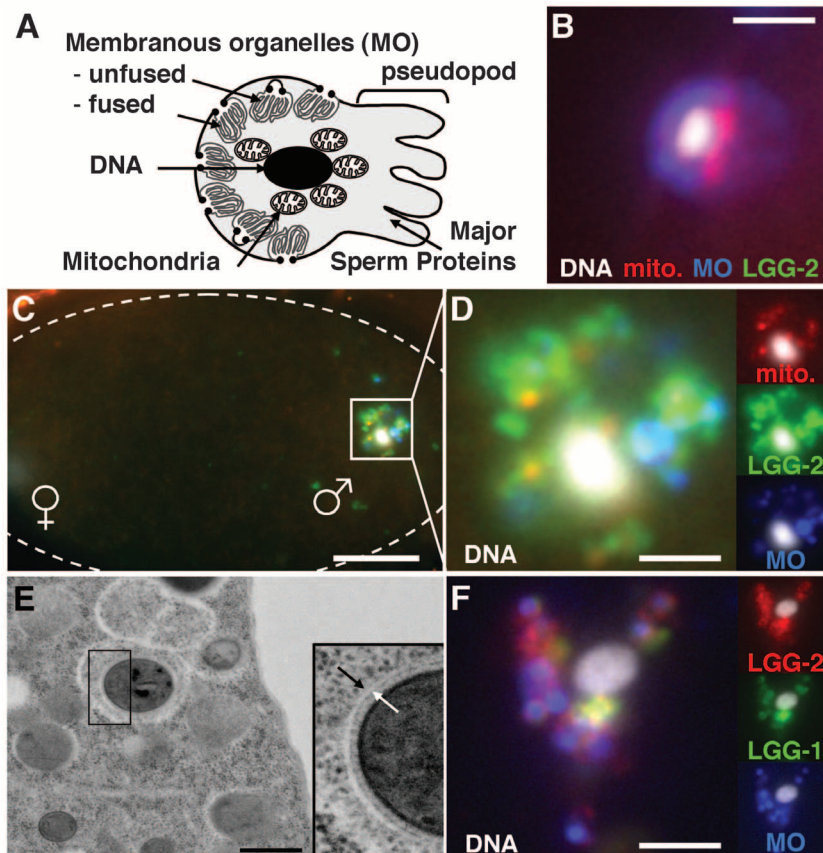
To test whether sperm entry could trigger an autophagic response, we stained embryos using antibodies for *C. elegans* Atg8/LC3 ubiquitin-like proteins: LGG-1 and LGG-2 (12). Both are recruited to the membrane of the autophagosomes and are good markers for macroautophagy (13). At 15 to 30 min postfertilization, prominent LGG-1- and LGG-2-positive structures appear next to the sperm DNA (Fig. 1, C, D, and F). Controls demonstrate that this staining is specific (fig. S1, E to H) and is absent in the mature spermatozoa (Fig. 1B). Time-lapse analysis using a GFP::LGG-2 revealed that these large structures are dynamic, tend to cluster around the sperm DNA, then segregate during the first mitotic divisions close to the cortex (movie S1) and disappear. LGG-1/LGG-2 signal surrounds MOs and paternal mitochondria, suggesting that these organelles are enwrapped by autophagosomes (Fig. 1, C, D, and F). This sperm-specific recruitment was confirmed in polyspermic *egg-4/5(RNAi)* embryos (RNAi, RNA interference) (14) in which LGG-2 autophagic structures are associated with MOs around each sperm DNA (fig. S1I). Electron microscopy analyses showed that after fertilization, mitochondria are found in double-membrane vesicles (Fig. 1E) in the cortical region,

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Fig. 1. The entry of spermatozoon organelles in the *C. elegans* oocyte at fertilization induces autophagy. (A) Schematic representation of a *C. elegans* spermatozoon. (B to D) Spermatozoon is devoid of LGG-2 (green) but transmits MOs (blue), CMXRos-labeled mitochondria (red), and DNA (white) (B) to the oocyte upon fertilization (C). (D) Magnified view of the sperm DNA (white) region. Twenty min after fertilization, organelles are associated with LGG-2 (green)-positive structures. Insets correspond to the split channels. (E) Transmission electron microscopy image of the cortical area of a two-cell-stage embryo revealing the presence of autophagosomes around mitochondria. Arrows indicate the limiting double membrane. (F) The two LC3 homologs, LGG-2 (red) and LGG-1 (green), are recruited to the MOs (blue) around sperm DNA (white). Scale bars: 2 μ m in (B), (D), and (F); 10 μ m in (C); 500 nm in (E). The dotted lines in (C) indicate the border of the embryo; male (σ) and female (φ) DNA are indicated.



demonstrating that mitophagy does occur at that time.

In mammals, ubiquitination of sperm mitochondria has been proposed to be the signal for their degradation (5). To test whether ubiquitination could serve as a mark for autophagy degradation in spermatozoa and early embryos, we used an antibody specific for ubiquitin conjugates in combination with either CMXRos-labeled sperm mitochondria or MO antibody (Fig. 2). We found that the ubiquitin antibody stains MOs, but not mitochondria, in mature spermatozoa and in newly fertilized embryos (Fig. 2, A to D). Using GFP::ubiquitin expressing worms, we observed its recruitment on structures around sperm DNA

within 3 min after fertilization, suggesting that sperm components are further ubiquitinated after they enter the oocyte (Fig. 2E and movie S2), and confirmed that sperm-inherited mitochondria are not ubiquitinated (fig. S3). Several types of ubiquitin modification exist, depending on the number and the branching of the ubiquitin moiety. Ubiquitin chains linked via their lysine-48 residue (K48-linked) can target the substrate for proteasome degradation, whereas K63-linked chains are associated with the autophagic pathway (15). Costaining MOs and mitochondria with either K63- or K48-specific antibodies (Fig. 2F) reveals that MOs are decorated only in the embryo with both K63 and K48 ubiquitin chain antibodies.

K63 ubiquitination of MOs precedes LGG-1/2 recruitment and autophagosome formation and is likely to be the signal for autophagy of these sperm organelles. The presence of the K48-linked chains signal and the 19S regulatory subunit of the proteasome around MOs (Fig. 2F) suggests that some proteasome activity could also be involved in MO degradation. But unlike the autophagy markers, the 19S subunit is inherited from the spermatozoa (Fig. 2F), suggesting a role before fertilization. Together, these data demonstrate that after fertilization, both MOs and mitochondria are associated with autophagosomal structures, although only MOs are ubiquitinated.

Fig. 2. Paternal inherited MOs are rapidly K63 and K48 ubiquitinated after fertilization, but mitochondria are not ubiquitinated. (A to D) MOs [red in (A and B)], but not mitochondria [red in (C and D)], are ubiquitinated (green) in the spermatozoon (A and C) and around sperm DNA after fertilization (B and D). (E) Still images of a confocal movie showing GFP::ubiquitin (green) recruitment around sperm chromatin (mCherry Histone-H2B, red) after fertilization, time indicated in minutes:seconds. (F) MOs (red) in the sperm DNA (blue) region are positive for K63 and K48 ubiquitination (green) but negative before fertilization. Mitochondria (red) remain negative (movie S3). The proteasome 19S subunit (green) is found in the spermatozoa, then around MOs but not mitochondria (red) after their entry. Scale bars: 2 μ m in (A) to (D) and (F); 5 μ m in (E).

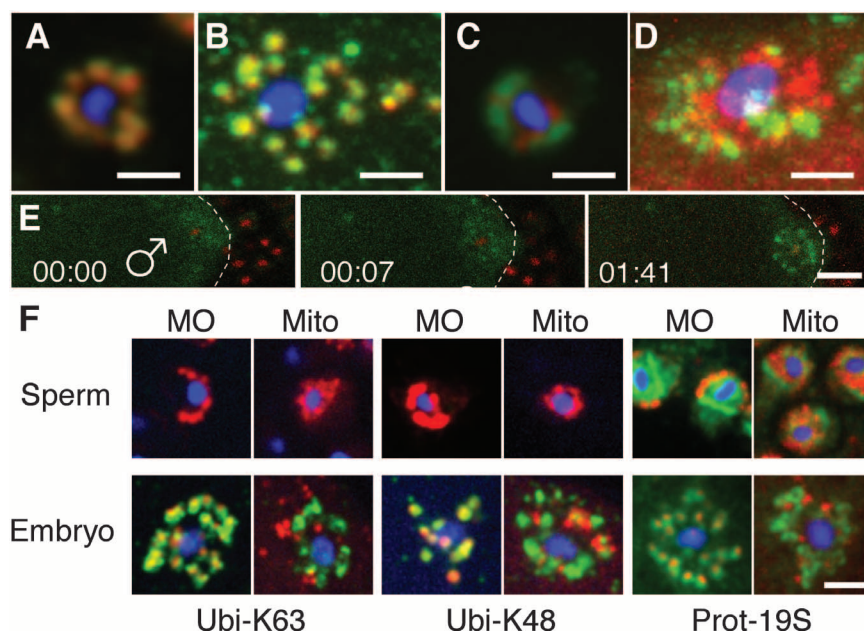


Fig. 3. Autophagy is required to degrade *C. elegans* spermatozoon inherited organelles after fertilization and prevent mitochondrial heteroplasmy. (A) Z-projections of confocal stacks. MOs (green) are absent in WT (Ctl) but remain in *lgg-1(tm3489)* 120-cell embryos. LGG-1 (red) is absent in the mutant embryo. Average number and SD of the remaining MO structures in 100/120-cell stage control ($n = 22$) and *lgg-1(tm3489)* ($n = 10$) embryos are indicated. (B and C) Paternal mitochondria and their DNA persist in autophagy-defective embryos. (B) Paternal mitochondria visualized by CMXRos mitotracker (green) are absent at the 100/120-cell stage in controls ($n = 31$), whereas they persist in *lgg-1+2(RNAi)* ($n = 23$) embryos. Average numbers of the remaining CMXRos and SDs are indicated below. Scale bars: 10 μ m. (C) Interfering with autophagy maintains paternal mitochondrial heteroplasmy. Heteroplasmic males carrying both deleted (*UaDf5*) and WT mtDNA were crossed with control or *lgg-1+2(RNAi)* hermaphrodites, and their embryos progeny were tested by PCR for the presence of WT and *UaDf5* mtDNA. Male *UaDf5* mtDNA is detected in autophagy-deficient embryos, but not in control, indicating a heteroplasmic state.

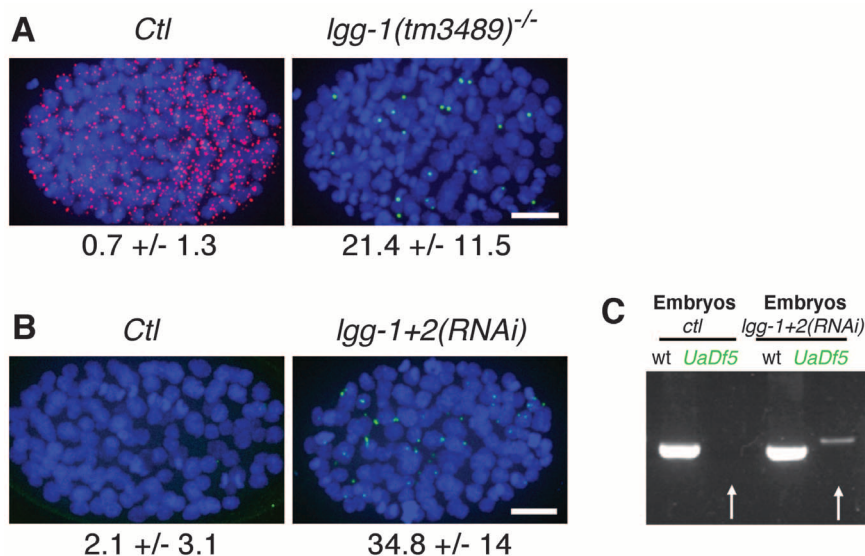
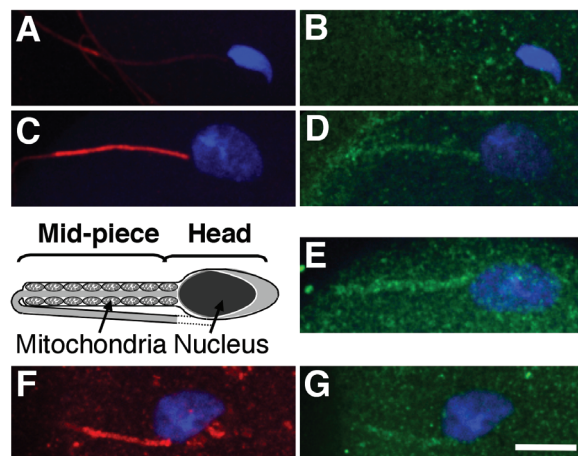


Fig. 4. Fertilization of the mouse oocyte induces recruitment of autophagy markers around the flagellum midpiece of the spermatozoon. The scheme represents the morphology of mouse spermatozoon with only a part of the flagellum. (A and C) Maximum intensity projection of Z-stacks of confocal images of the entire volume of the oocyte or of three consecutive images (E) or single confocal images (B, D, F, and G). (A and B) Unfertilized or (C to G) fertilized mouse oocytes stained for sperm DNA (blue), immunostained for ubiquitin (red) (A and C), LC3 (green) (B and D), GABARAP (green) (E), and K63-linked ubiquitin (F) together with P62 (G). The staining appears in the midpiece associated with the partially decondensed male pronucleus (C to G), whereas only faint (A) or no (B) signals are visible in the midpiece of external spermatozoa (A and B). Scale bar: 5 μ m.



To confirm that the formation of autophagic structures is involved in the degradation of sperm organelles, we used *lgg-1(tm3489)* mutant and *lgg-1+2(RNAi)* to inactivate autophagosome formation in the oocyte. In control embryos the number of MOs rapidly decreases during the first cell divisions, whereas in mutant and RNAi embryos, MOs are still present at 100/120-cell stage (Fig. 3A and fig. S4A). To confirm that the persistence of MOs is due to the reduction of the autophagy activity we depleted two other proteins involved in autophagic pathway: ATG-7, the E1-like activating enzyme required for ATG8/LC3/LGG-1 activation (16), and the small guanosine triphosphatase Rab7 involved in the fusion of autophagosomes with lysosomes (17). Both depletions induce an alteration of LGG-2 signal and a persistence of MOs (fig. S2, C and D) supporting the notion that MOs are degraded by macro-autophagy.

We then analyzed whether paternal mitochondria, which are not ubiquitinated but are closely associated with LGG-1/LGG-2 autophagic structures in the one-cell embryo, are also degraded by autophagy. We crossed ANT-1.1::GFP- and CMXRos-labeled males with control and *lgg-1+2(RNAi)* hermaphrodites, and we analyzed the presence of paternal mitochondria in their progeny. Whereas in control embryos paternal mitochondria are not detected from the 8/10-cell stage, they are maintained in *lgg-1+2(RNAi)* embryos (Fig. 3B and fig. S4B). The persistence of paternal mitochondria in defective autophagy conditions suggests that paternal mtDNA could also persist in embryo. To track paternal mtDNA in the fertilized embryos, we used males from the LB138 strain, which contains both WT and deleted mtDNA (*Uad5*) (4). When LB138 males were crossed with control worms, we were not able to amplify the paternal mtDNA in the progeny by polymerase chain reaction (PCR) (Fig. 3C and fig. S4C), confirming that it is efficiently degraded in WT embryos. However, in *lgg-1+2(RNAi)* embryos, the paternally inherited mito-

chondrial DNA was detected in 2- to 6-hour-old embryos (80 to 500 cells) (Fig. 3C). Taken together, these data indicate that after fertilization, paternal mitochondria and their genomes are quickly degraded by autophagy and that inactivation of autophagy results in heteroplasmy in the embryo.

Our findings in *C. elegans* demonstrate that spermatozoon fertilization of the oocyte triggers a selective autophagic process that recognizes and degrades paternally inherited organelles but not cytoplasmic MSPs. In mammals, ubiquitinated sperm mitochondria, localized in the midpiece of the flagellum, enter the oocyte after gamete fusion and are degraded by an unknown mechanism (5). Mouse oocyte fertilization induces two essential waves of autophagy several hours after sperm entry that have been hypothesized to participate in the degradation of maternal material (9). To test whether autophagy of spermatozoon components could be conserved in mammals, we costained fertilized mouse oocytes with anti-ubiquitin and anti-LC3 antibodies. As reported (5), ubiquitinated mitochondria are detected in the midpiece of the sperm before [21 ubiquitinated mitochondria out of 21 spermatozoa before fertilization (21/21)] and, even brighter, after fertilization (23/24) (Fig. 4, A and C, and fig. S5, A, B, E, and F). We observed that the midpiece was stained with the LC3 antibody after fertilization (13/13) and not in spermatozoa outside the oocyte (3/3) (Fig. 4, B and D, and fig. S5, C, D, G, and H). We further confirmed that autophagy hallmarks label the mid-piece of spermatozoa using antibodies against gamma-aminobutyric acid receptor-associated protein (GABARAP), another ATG8 homolog (9/9), and Ubi-K63 and P62 (42/49), an autophagic adaptor (Fig. 4, E, F, and G, and fig. S5, I and J; K, L, O, and P; M, N, Q, and R, respectively). Recruitment of autophagosomal markers in the vicinity of ubiquitinated mitochondria suggests that this autophagy could also exist in mammals and be involved in the degradation of sperm-inherited mito-

chondria. It is notable that, in *C. elegans*, the paternal mitochondria are not ubiquitinated, which indicates that this autophagy is able to degrade both ubiquitinated and nonubiquitinated substrates. It is possible that the mitochondria are incorporated into autophagosomes due to their close association with ubiquitinated MOs. Deciphering the signaling cascade that triggers spermatozoon autophagy and the mechanisms underlying the selectivity of this response would help us to better understand this process and could affect human medically assisted reproduction or animal cloning.

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Figs. S1 to S5

References

Movies S1 to S3

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